1. What are data types in Java? Explain the different types of data types.
2. What is the difference between **primitive and non-primitive data types** in Java?
3. What are the **8 primitive data types** in Java? Explain each with its size and default value.
4. What is the difference between **int, float, double, and long**?
5. What is the difference between **char and String** in Java?
6. What is a variable in Java? Explain **local, instance, and static variables**.
7. What is the difference between **local variable and instance variable**?
8. What is the difference between **declaration, initialization, and assignment** of a variable?
9. What are **literals** in Java? Explain different types of literals with examples.
10. What are operators in Java? Explain the different types of operators.
11. What is the difference between **`=`, `==` and `equals()`**?
12. What is the difference between **`&&` and `&`** operators?
13. What is the difference between **`||` and `|`** operators?
14. What is the difference between **pre-increment (`++a`) and post-increment (`a++`)**?
15. What is **type casting** in Java? Why is it required?
16. What is the difference between **widening and narrowing type casting**?
17. What happens when you convert a **double to int**? Explain with an example.
18. What is the difference between **if-else and switch** statements? When would you use each?
19. What is the difference between **for, while, and do-while loops**?
20. What is the difference between **break, continue, and return** statements?

APTITUDE TEST – SET 3

Percentage, Profit & Loss, Ratio & Proportion | 20 Questions

1. In a company, 40% of employees are women. If 15% of the women and 20% of the men are working remotely, what percentage of all employees are working remotely?

A) 18% B) 20% C) 22% D) 24%

2. A number is 30% of another number. The smaller number is increased by 40%, while the larger number is decreased by 10%. What is the new ratio of smaller to larger?

A) 2:3 B) 7:15 C) 14:27 D) 3:5

3. A company's revenue rises by 25%, but its profit margin falls from 20% to 16%. If the original revenue was ₹8 lakh, what is the percentage change in profit?

A) 0% B) 5% decrease C) 10% decrease D) 20% increase

4. A student needs 40% to pass an examination. He scores 216 marks and fails by 24 marks. What percentage would he have scored if he had obtained 36 additional marks?

A) 40% B) 42% C) 44% D) 46%

5. A shop reduces the price of an item by 12.5%. With the same amount of money, a customer can buy 4 more items. If the original quantity purchased was 28 items, how many items can be purchased now?

A) 30 B) 32 C) 34 D) 36

6. The length of a rectangle is increased by 20% and its breadth is reduced by 10%. What is the percentage change in its area?

A) 8% increase B) 10% increase C) 8% decrease D) 10% decrease

7. A number is first reduced by 25%. By what percentage of the reduced number must it be increased to become 10% greater than the original number?

A) 40% B) 45% C) 46 2/3% D) 50%

8. A trader buys 12 identical articles for ₹9,600. He sells 8 of them at ₹1,000 each and the remaining 4 at ₹800 each. What is his overall profit percentage?

A) 12.5% B) 15% C) 16 2/3% D) 20%

9. A shopkeeper sells an article at 12% profit. If he had bought it for 10% less and sold it for ₹40 less, his profit would have been 20%. Find the original cost price.

A) ₹800 B) ₹900 C) ₹1,000 D) ₹1,200

10. A trader marks an article so that after a discount of 20%, he earns 25% profit. If the cost price is ₹1,600, what is the marked price?

A) ₹2,400 B) ₹2,500 C) ₹2,600 D) ₹2,800

11. A seller incurs a loss of 15% on an article. If both the selling price and cost price were ₹120 higher, the loss would be 10%. Find the original cost price.

A) ₹2,000 B) ₹2,400 C) ₹2,800 D) ₹3,000

12. A dealer buys an article at 20% below its marked price and sells it at 10% below the marked price. What is his profit percentage on the purchase price?

A) 10% B) 12.5% C) 15% D) 18%

13. An article is sold for ₹2,640 after two successive discounts of 10% and 20%. What was its marked price?

A) ₹3,500 B) ₹3,600 C) ₹3,666 2/3 D) ₹3,750

14. A businessman earns 18% profit by selling an article for ₹2,360. If he wants to earn 25% profit on the same article, by how much should he increase the selling price?

A) ₹120 B) ₹130 C) ₹140 D) ₹150

15. Three numbers are in the ratio 2:3:5. If 12 is added to each, they become proportional to 4:5:7. Find the middle number originally.

A) 12 B) 18 C) 24 D) 30

16. A sum is divided between A and B in the ratio 7:5. If A gives ₹120 to B, their amounts become equal. What was the original sum?

A) ₹1,200 B) ₹1,320 C) ₹1,440 D) ₹1,560

17. The numbers of red, blue and green balls are in the ratio 3:4:5. If the total number of balls is 72, how many green balls are there?

A) 20 B) 24 C) 30 D) 36

18. A:B = 2:3 and B:C = 4:5. If A + B + C = 280, find C.

A) 100 B) 120 C) 140 D) 150

19. A can complete a task in 12 days and B can complete it in 18 days. If they work together, the ratio of A's share of the work to B's share is:

A) 2:3 B) 3:2 C) 4:3 D) 3:4

20. A mixture contains alcohol and water in the ratio 3:2. If 10 litres of water are added, the ratio becomes 3:4. What was the original quantity of the mixture?

A) 20 L B) 25 L C) 30 L D) 35 L

Puzzles

1. A prisoner is shown two doors. One door leads to freedom and the other to danger. There are two guards: one always tells the truth and the other always lies. You do not know which guard is which. You may ask one guard one question. What question guarantees that you choose the door to freedom?

2. You have a 7-minute hourglass and an 11-minute hourglass. Neither has markings between the start and finish. How can you measure exactly 15 minutes?

3. Three people check into a hotel room that costs ₹30. They each pay ₹10. Later, the manager realizes the room should cost ₹25 and sends ₹5 back with the bellboy. The bellboy keeps ₹2 and gives ₹1 back to each person. Each person has now paid ₹9, so 3 × ₹9 = ₹27, and the bellboy has ₹2, making ₹29. Where is the missing ₹1?

4. A person has a 4-minute timer and a 7-minute timer. Both can be started at any time, but they cannot be stopped midway. How can the person measure exactly 9 minutes?

5. You have 1000 bottles of water, and exactly one bottle contains poison. The poison is fatal, but it takes exactly 24 hours to show its effect. You have 10 test subjects and only 24 hours to identify the poisoned bottle. How can you do it?